

An Internship Report
Of
Data Science Master

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B-Tech (CSE)



By

Roll Number of Student:- 2410031737

Name of Student: Saumya Kashyap

Batch: 2026-27

CANDIDATE'S DECLARATION

I, Saumya Kashyap, do hereby declare that the internship report titled “Data Science Master” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: Saumya Kashyap

Roll No.: 2410031737

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the **AICTE – EduSkills Virtual Internship Program team and EduSkills Academy** for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of Data Science Master. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:

Student Name: Saumya Kashyap
Roll No.: **2410031737**

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3. INTERNSHIP COMPLETION CERTIFICATE



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MINISTRY OF
EDUCATION



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NATIONAL
INTERNSHIP
PORTAL



EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

Saumya Kashyap

IILM University, Greater Noida

has successfully completed the 8-weeks

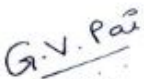
Data Science Master Virtual Internship

During June - August 2026

May this Internship learning propel you toward a bright and successful career.

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Certificate ID : 40308bdd9c2621ea6afa

Student ID: STU6a465b910419717





GRADE: O (Outstanding): 90-100 | E (Excellent): 80-89 | S (Satisfactory): 70-79 | P (Pass): 60-69 | F (Fail): Below 60

Internship Offer Letter



Ref No: C17Y2026M071492462

Internship Offer Letter

Student Name : Saumya Kashyap
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : Siemens Data Science Master
Internship Duration : June 2026 - August 2026
AICTE Student ID : STU6a465b91041971782995857

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning.
Week 2	Data Engineering Professional	Learn to access, load, transform, and calculate data.
Week 3	Machine Learning Professional	Build foundational machine learning skills.
Week 4	Data Preparation Master	Master advanced data handling and automation.
Week 5	Machine Learning Master	Learn advanced machine learning techniques.
Week 6	Applications & Use Cases Master	Deploy and monitor ML models.
Week 7	Platform Administration Master	Configure and manage RapidMiner AI Hub.
Week 8	Platform Administration Master	Enterprise-level platform operations.
Final Credential Validation / Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.



4. PROJECT DESCRIPTION

4.1 Introduction

Data Science is an interdisciplinary field that uses data, statistical methods, programming, machine learning, and analytical techniques to extract useful information and support decision-making. In modern organizations, data science plays an important role in identifying patterns, predicting future outcomes, automating processes, and improving business operations.

The AICTE–EduSkills Virtual Internship was undertaken to gain practical exposure to important concepts related to Data Science and Machine Learning. According to the internship offer letter, the program followed a structured eight-week learning plan with weekly assessments, assignments, project documentation, and a final assessment.

The internship domain assigned was **Siemens Data Science Master**. The learning structure covered business problem identification, data engineering, machine learning, data preparation, advanced machine learning techniques, deployment and monitoring of machine learning models, and RapidMiner AI Hub administration.

4.2 Organization Profile

EduSkills Academy is an ISO 9001:2015 and ISO 20000-1:2018 certified organization working under the theme “Nation Building Through Skills”. EduSkills partners with the All India Council for Technical Education (AICTE) to run the AICTE – EduSkills Virtual Internship Program under the patronage of the AICTE National Internship Portal. The organization collaborates with leading technology companies and academic institutions to design industry-aligned virtual internship curricula that help engineering students strengthen their employability and gain practical exposure to emerging technology domains such as Artificial Intelligence, Cloud Computing, Cybersecurity, and MLOps, among others.

4.3 Problem Statement

In today's organizations, large amounts of data are generated from different sources. However, raw data by itself does not provide meaningful insights unless it is properly collected, prepared, analyzed, and converted into useful information.

The major problem addressed through this internship was the need to develop practical knowledge of the **end-to-end data science process**. This includes understanding how to:

- Identify business problems that can be addressed using data science and machine learning.
- Access, load, transform, and calculate data.
- Prepare raw data for analysis and machine learning.
- Apply suitable machine learning techniques.
- Build and evaluate machine learning solutions.
- Deploy and monitor machine learning models.
- Understand the administration and management of a data science platform.

4.4 Project Objectives

The objective of the internship was to develop practical knowledge of **Data Science and Machine Learning** by understanding the complete data science workflow, including data engineering, data preparation, machine learning, model deployment, monitoring, and platform administration. The internship aimed to strengthen **technical, analytical, and problem-solving skills** through structured learning and assessments.

4.5 Scope of the Project

The scope of the internship project covers the fundamental and practical aspects of Data Science and Machine Learning. The internship was structured to provide learning across the complete data science workflow.

The major areas within the scope are:

- **Business Problem Identification**

Understanding business requirements and identifying problems that can be solved using Artificial Intelligence and Machine Learning.

- **Data Engineering**

Learning how data can be accessed, loaded, transformed, and processed for further analysis.

- **Machine Learning**

Understanding the fundamentals of machine learning and learning how machine learning techniques can be applied to data-driven problems.

- **Data Preparation**

Understanding the importance of preparing and transforming data before using it for machine learning.

- **Machine Learning Mastery**

Developing deeper knowledge of machine learning techniques and their practical applications.

- **Model Deployment and Monitoring**

Understanding how machine learning models can be deployed and monitored after development.

- **Platform Administration**

Gaining exposure to RapidMiner AI Hub configuration, management, and enterprise-level platform operations.

4.6 Technologies and Tools Used

Technology / Tool	Purpose
Data Science	Analysis of data and extraction of meaningful insights
Machine Learning	Development of predictive and analytical solutions
Data Engineering	Accessing, loading, transforming, and processing data
Data Preparation	Cleaning and preparing data for analysis and machine learning
RapidMiner AI Hub	Platform for managing and operating data science and machine learning workflows
Artificial Intelligence	Identification and application of AI-based solutions to suitable business problems

4.7 System Architecture

Architecture Components

1. Business Problem Identification

The process begins by identifying a suitable business problem where Artificial Intelligence or Machine Learning can provide a solution.

2. Data Access and Collection

Relevant data is accessed and loaded from appropriate sources for further processing.

3. Data Engineering

The collected data is transformed and processed into a suitable format for analysis.

4. Data Preparation

The data is prepared for machine learning by performing the necessary preprocessing and transformations.

5. Machine Learning Model Development

Suitable machine learning techniques are applied to the prepared data to develop a model.

6. Model Evaluation

The developed model is evaluated to understand its performance and suitability for the identified problem.

7. Deployment

A suitable machine learning model can then be deployed for practical use.

8. Monitoring

The deployed model can be monitored to ensure that it continues to perform appropriately.

9. Platform Administration

RapidMiner AI Hub provides an environment for managing and administering enterprise-level data science and machine learning operations.

4.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning, hands-on labs, weekly assessments, and project submissions, as summarized in the table below.

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learned to identify business problems that are suitable for Artificial Intelligence and Machine Learning solutions.
Week 2	Data Engineering Professional	Learned how to access, load, transform, and calculate data for further analysis and processing.

Week	Module	Description
Week 3	Machine Learning Professional	Learned the fundamentals and basic concepts of Machine Learning and its applications.
Week 4	Data Preparation Master	Learned advanced data handling, preparation, transformation, and automation techniques.
Week 5	Machine Learning Master	Learned advanced Machine Learning techniques and their application to data-driven problems.
Week 6	Applications & Use Cases Master	Learned how to deploy and monitor Machine Learning models for practical applications.
Week 7	Platform Administration Master	Learned to configure and manage the RapidMiner AI Hub platform.
Week 8	Platform Administration Master	Learned about enterprise-level platform operations and administration of RapidMiner AI Hub.

Each week concluded with a weekly assessment to evaluate conceptual understanding, and selected modules required submission of project documentation and assignments. In the final week, a Final Assessment Test was conducted, and the overall grade was based on performance across weekly assessments, project submissions, and the final test.

4.9 Expected Outcomes

The expected outcome of the internship was to develop practical, industry-oriented knowledge of Data Science and Machine Learning.

After completing the internship, the learner is expected to be able to:

- Understand the role of Data Science in solving business problems.
- Identify suitable problems for Artificial Intelligence and Machine Learning.
- Understand the basic data engineering workflow.
- Work with data through loading, transformation, and preparation.
- Understand fundamental and advanced machine learning concepts.

- Understand the process of deploying and monitoring machine learning models.
- Gain introductory knowledge of RapidMiner AI Hub administration.
- Apply analytical and problem-solving skills to data-driven problems.
- Understand the overall lifecycle of a machine learning solution.
- Develop a foundation for further study and professional work in Data Science and Machine Learning.

The internship is therefore expected to strengthen both technical knowledge and practical understanding of the data science lifecycle.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter and the Internship Completion Certificate issued by EduSkills Academy, confirming successful completion of the 10-weeks AICTE – EduSkills Virtual Internship Program in Data Science Master, are attached in Chapter 3 of this report.



EduSkills Fo... 2 Aug
to me ▾



Dear **Saumya Kashyap**,

 **Congratulations!**

You have successfully completed **10 weeks Siemens Data Science Master** internship with EduSkills Foundation. Your certificate is ready — please log in to your profile and download it.



Steps to Download Your Certificate:

1. Login at: <https://aicte-internship.eduskillsfoundation.org/>
2. Go to the **Certificate Section** from your sidebar
3. Click **Download Certificate**

Please post your certificate on your **LinkedIn**

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